HAYA ALSHAYJI

Ph.D. Candidate – Industrial & Manufacturing Engineering

301 Leonhard Building, University Park, PA 16802 | hka5222@psu.edu | 714-930-3176

Comprehensive examination passed ● Dissertation phase ● Expected defense: Summer/Fall 2026

DISSERTATION RESEARCH

Promoting Equity in Mental Health and Addressing Diseases of Despair—a three-paper dissertation using a TriNetX Diamond Network EHR cohort of nearly 500,000 patients linked to 3-digit ZIP-level social-determinant data. Paper 1: completed multilevel logistic regression of social determinants of diseases of despair. Paper 2: interpretable ML risk classification (LR, RF, XGBoost) with SHAP explanation and a stratified fairness audit. Paper 3: Decision-analytic cost-effectiveness analysis of community-based interventions from a Medicaid-payer perspective—the methodological core of a submitted CHERISH pilot grant proposal.

Committee: Paul M. Griffin, Ph.D.; Soundar Kumara, Ph.D.; Christopher Dancy, Ph.D.; Vasant Honavar, *Ph.D.*

EDUCATION

Ph.D., Industrial and Manufacturing Engineering *2020–Present*

The Pennsylvania State University, University Park, PA

Cumulative GPA: 3.56 / 4.00 | Comprehensive exam: passed

Selected coursework: Engineering Analytics; Data Mining: Techniques and Applications; Regression Methods; Assessing the Value of Health Interventions; Healthcare Systems Engineering (audited); Principles of Causal Inference (audited).

M.S., Industrial and Systems Engineering *2017–2019*

San Jose State University, San Jose, CA

GPA: 3.67 / 4.00

Selected coursework: Analytics for Systems Engineering; Healthcare and Service Systems Engineering.

B.S., Industrial and Manufacturing Systems Engineering *2007–2012*

Kuwait University, Kuwait

Capstone advisor: Ali Allahverdi, Ph.D. — Outstanding IMSE Senior Project Award.

HONORS & AWARDS

- Selected for Full Scholarship — Data and AI Intensive Research with Rigor and Reproducibility (DAIR3) Program, University of Michigan — 2026.
- Finalist, 3rd Coast AI for Health Bowl, Institute for Augmented Intelligence in Medicine (I.AIM), Northwestern University — 2023.
- STAR Fellow, Industrial & Systems Engineering Department, San Jose State University — 2018–19.
- Davidson Student Scholar, College of Engineering, San Jose State University — 2018–19.
- Graduate Scholarship for Master's and Doctoral Studies, Public Authority of Applied Education and Training (PAAET), Kuwait — 2017.
- Outstanding IMSE Senior Project, Kuwait University — 2012.
- Member, Tau Beta Pi Engineering Honor Society.

GRANTS & SUBMITTED PROPOSALS

- CHERISH Pilot Grant Proposal — "Cost-Effectiveness of Mobile Clinic and Counseling Interventions for Diseases of Despair Among Medicaid-Enrolled Populations Identified by Machine Learning Risk Stratification." Submitted 2026. Mentor: Bruce Schackman, Ph.D. (Weill Cornell Medicine).

RESEARCH EXPERIENCE

Graduate Researcher — LISA Lab & AI Research Lab, Penn State, *2023–Present*

- Constructed and analyzed a TriNetX Diamond Network EHR cohort (566,752 patients) linked to 3-digit ZIP-level Social Vulnerability Index data; completed multilevel logistic regression in R (glmmTMB) for the dissertation's Paper 1.
- Developing an interpretable ML risk-classification pipeline (Logistic Regression, Random Forest, and XGBoost) with SHAP-based explanation and a stratified fairness audit by sex, race, ethnicity, age, and insurance for Paper 2.
- Designing a decision-analytic cost-effectiveness model with probabilistic sensitivity analysis for community-based interventions from a Medicaid-payer perspective—the core of an active CHERISH Pilot Grant proposal (Paper 3).
- Leading parallel work on an operational mapping of food insecurity and health burden in Pennsylvania—a county-level service-gap index integrating publicly available data on food insecurity, poverty, chronic disease, and preventable hospitalizations to guide equitable mobile-pantry deployment (presented at the PA Community and Public Health Conference 2026).
- Co-supervised by Drs. Paul Griffin, Soundar Kumara, and Vasant Honavar.

Graduate Research Assistant — THiCC Lab, Penn State, *2024*

- Led a study on critical AI literacy for the Multicultural Engineering Summer Bridge program, resulting in a peer-reviewed paper presented at ASEE 2025 (Montreal).
- Designed AI-integrated learning materials and mentored first-year engineering students; contributed analyses to human–AI interaction projects.

Team Member — 3rd Coast AI for Health Bowl, Northwestern (I.AIM), *2022–2023*

- Co-developed "NittanyNav," an AI tool addressing healthcare disparities; advanced through semifinals to the national finals.

Graduate Research Assistant — San Jose State University *2018–2019*

- Primary researcher on rule-based smart-contract design in supply chains (Dr. Hongrui Liu); IISE 2019 conference paper.
- Conducted a systematic literature review on training decay and task categorization in the Human Factors USER lab (Dr. Dan Nathan-Roberts).

PEER-REVIEWED PUBLICATIONS

- **Alshayji, H.**, Workman, D., Dulam, S., Griggs, L. A., Zor, D., & Dancy, C. L. (2025, June). Experiences with Using an LLM-based Chatbot for a Multicultural Engineering Program Orientation. ASEE Annual Conference & Exposition, Montreal, QC. https://doi.org/10.18260/1-2--56487
- **Alshayji, H.**, & Liu, H. (2019). Rule-based Smart Contract Design in Supply Chain. Proceedings of the IISE Annual Conference, Orlando, FL.

MANUSCRIPTS UNDER REVIEW

- **Alshayji, H.**, Griffin, P. M., Kumara, S. Multilevel Effects of Clinical and Social Determinants of Health on Diseases of Despair: A Nationwide Retrospective Study. Submitted to BMJ Open.
- Alkhadhr, S. (corresponding author), Almonaies, A., & **Alshayji, H.** PEURLNet: A Multimodal Neural Framework for Page, Email, and URL Phishing Detection. Submitted to High-Confidence Computing (Elsevier).

SELECTED PRESENTATIONS

- Abu Aridha, S., Alkhadhr, S., & **Alshayji, H.** (2026, May). AI-Driven Resilience in Healthcare Systems: Integrating Bed-Occupancy Forecasting and Supply Chain Optimization for Critical Medical Resources. Accepted, 2026 IISE Annual Conference & Expo, Bioprocess & Clinical Healthcare Optimization session (Data Analytics and Information Systems track), Arlington, TX.
- **Alshayji, H.** (2026, April). Operational Mapping of Food-Insecurity and Health Burden in Pennsylvania: A Pilot Framework for Data-Driven Public-Health Decision-Making. 2026 Pennsylvania Community and Public Health Conference, Lancaster, PA.

- **Alshayji, H.** (2024, November). Enhancing Diagnosis and Prognosis of Diseases of Despair: An Attention Network Approach. INFORMS Annual Meeting.
- **Alshayji, H.**, & Vanness, D. (2024, Fall). Understanding the Effect of Economic Factors on Life Expectancy in Asian Countries. ICDS Symposium, Penn State.
- **Alshayji, H.**, et al. (2023, May). NittanyNav: 3rd Coast AI for Health Bowl Finals, Northwestern University.
- **Alshayji, H.**, Ventura, M., & Padmanabhan, A. (2021, November). Development of a COVID-19 Vaccination Plan Considering Vaccine Availability and Temporary Delivery Centers. INFORMS Annual Meeting.

LEADERSHIP, SERVICE & MENTORING

- Mentor, Multicultural Engineering Summer Bridge Program, Penn State — 2024.
- Member, Institute for Operations Research and the Management Sciences (INFORMS).
- Member, Institute of Industrial and Systems Engineers (IISE).

TECHNICAL SKILLS

Programming — R / RStudio, Python, C++.

Tools — Minitab, Simio, Arena, LINDO, LaTeX, MS Office, Adobe Suite.

Methods — multilevel/hierarchical modeling, machine learning, causal inference, health-economic evaluation, and large-scale EHR data engineering.

Languages — Arabic (native), English (fluent).

RESEARCH INTERESTS

Healthcare data analytics; diseases of despair; social determinants of health; machine learning and AI applications in healthcare; multilevel and hierarchical modeling; health equity and disparities; interpretable and responsible AI; health-economic evaluation; population health analytics.

REFERENCES

Paul M. Griffin, Ph.D.
Lucas Professor; Director, Penn State Consortium on Substance Use & Addiction
The Pennsylvania State University | *pmg14@psu.edu* | *814-865-7601*

Soundar Kumara, Ph.D.
Allen E. Pearce/Allen M. Pearce Professor; Director, Center for Applications of AI & ML to Industry
The Pennsylvania State University | *u1o@psu.edu* | *814-863-2359*

Cindy Reed, Ph.D.
Assistant Director, Student Research and Graduate Equity; Associate Teaching Professor
The Pennsylvania State University | *chr10@psu.edu* | *814-863-5322*